DX-IMU6 6-Axis Motion Sensor - Product Datasheet Rev. 1.2

FICTIONAL DEVICE - sample document for the genai-engineering-assistant project. It does not describe a real product.

1 Overview

The DX-IMU6 combines a 3-axis accelerometer and a 3-axis gyroscope with a 16-bit ADC per axis, a digital low-pass filter, a temperature sensor and an I2C interface (address 0x68, or 0x69 with AD0 high). Typical applications: robot and drone attitude estimation, model-car odometry, vibration monitoring.

Features: accelerometer ranges ±2, ±4, ±8, ±16 g; gyroscope ranges ±250, ±500, ±1000, ±2000 °/s; 1 kHz internal sample rate; data-ready interrupt; 3.9 mA in normal mode, 5 µA in sleep.

2 Pin description

Pin	Name	Description
1	VDDIO	I/O supply
2	SCL	I2C clock
3	SDA	I2C data
4	AD0	I2C address bit 0
5	INT	Interrupt output, push-pull
6	GND	Ground
7	REGOUT	Regulator filter capacitor, 100 nF to GND
8	VDD	Supply

Place a 100 nF decoupling capacitor close to VDD and to VDDIO. SDA and SCL need pull-up resistors to VDDIO (4.7 kΩ for 100 kHz, 2.2 kΩ for 400 kHz).

3 Electrical characteristics

3.1 Absolute maximum ratings (stress beyond these values may cause permanent damage)

Parameter	Symbol	Min	Max	Unit	Notes
Supply voltage	VDD	-0.3	4	V	
Storage temperature	TSTG	-40	125	°C	
Mechanical shock			10000	g	0.2 ms

3.2 Operating conditions (TA = 25 °C unless noted)

Parameter	Symbol	Min	Typ	Max	Unit	Conditions
Supply voltage	VDD	2.375	3.3	3.46	V	
I/O supply voltage	VDDIO	1.71	1.8	3.46	V	VDDIO must not exceed VDD
Supply current, normal mode	IDD		3.9		mA	accel and gyro on
Supply current, sleep	IDD_SLP		5		µA	
Operating temperature	TA	-40		85	°C	
I2C clock frequency	fSCL			400	kHz	

4 I2C interface timing

Parameter	Symbol	Min	Max	Unit
SCL clock frequency	fSCL		400	kHz
Start condition hold time	tHD;STA	0.6		µs
SCL low period	tLOW	1.3		µs
SCL high period	tHIGH	0.6		µs
Data setup time	tSU;DAT	100		ns
Bus free time between stop and start	tBUF	1.3		µs

Burst reads auto-increment the register address. Read ACCEL_XOUT_H first: the low byte is latched when the high byte is read.

5 Register map

Addr	Register	Access	Reset	Description
0x19	SMPLRT_DIV	RW	0x00	Sample rate divider: rate = 1 kHz / (1 + SMPLRT_DIV)
0x1A	CONFIG	RW	0x00	Digital low-pass filter configuration
0x1B	GYRO_CONFIG	RW	0x00	Gyroscope configuration
0x1C	ACCEL_CONFIG	RW	0x00	Accelerometer configuration
0x38	INT_ENABLE	RW	0x00	Interrupt enable
0x3A	INT_STATUS	R	0x00	Interrupt status, cleared on read
0x3B	ACCEL_XOUT_H	R	0x00	Accelerometer X, high byte
0x3C	ACCEL_XOUT_L	R	0x00	Accelerometer X, low byte
0x41	TEMP_OUT_H	R	0x00	Temperature, high byte
0x42	TEMP_OUT_L	R	0x00	Temperature, low byte
0x43	GYRO_XOUT_H	R	0x00	Gyroscope X, high byte
0x44	GYRO_XOUT_L	R	0x00	Gyroscope X, low byte
0x6B	PWR_MGMT_1	RW	0x40	Power management 1
0x75	WHO_AM_I	R	0x6C	Device identity, always reads 0x6C

6 Register descriptions

6.x SMPLRT_DIV (address 0x19, RW, reset value 0x00)

Bits	Field	Function
7:0	SMPLRT_DIV	Divider value

6.x CONFIG (address 0x1A, RW, reset value 0x00)

Bits	Field	Function
5:3	EXT_SYNC	External frame sync
2:0	DLPF_CFG	Low-pass filter bandwidth

6.x GYRO_CONFIG (address 0x1B, RW, reset value 0x00)

Bits	Field	Function
7	XG_ST	X gyro self-test
6	YG_ST	Y gyro self-test
5	ZG_ST	Z gyro self-test
4:3	FS_SEL	Gyro full-scale range

FS_SEL	Full-scale range	Sensitivity
0	±250 °/s	131.1 LSB/(°/s)
1	±500 °/s	65.5 LSB/(°/s)
2	±1000 °/s	32.8 LSB/(°/s)
3	±2000 °/s	16.4 LSB/(°/s)

6.x ACCEL_CONFIG (address 0x1C, RW, reset value 0x00)

Bits	Field	Function
7	XA_ST	X accel self-test
6	YA_ST	Y accel self-test
5	ZA_ST	Z accel self-test
4:3	AFS_SEL	Accel full-scale range

AFS_SEL	Full-scale range	Sensitivity
0	±2 g	16384 LSB/g
1	±4 g	8192 LSB/g
2	±8 g	4096 LSB/g
3	±16 g	2048 LSB/g

6.x INT_ENABLE (address 0x38, RW, reset value 0x00)

Bits	Field	Function
4	FIFO_OFLOW_EN	FIFO overflow interrupt enable
0	DATA_RDY_EN	Data ready interrupt enable

6.x INT_STATUS (address 0x3A, R, reset value 0x00)

Bits	Field	Function
4	FIFO_OFLOW_INT	FIFO overflow occurred
0	DATA_RDY_INT	New data ready

6.x PWR_MGMT_1 (address 0x6B, RW, reset value 0x40)

Bits	Field	Function
7	DEVICE_RESET	Reset all registers
6	SLEEP	Sleep mode
5	CYCLE	Cycle mode
3	TEMP_DIS	Disable temperature sensor
2:0	CLKSEL	Clock source

Output data is 16-bit two's complement, high byte first. Temperature in °C = TEMP_OUT / 340 + 36.53.

7 Ordering information and revision history

Part number	Package	Temperature range
DX-IMU6-QFN	QFN-24, 4 x 4 mm	-40 to 85 °C

Revision	Date	Change
1.0	2026-03-01	First release
1.2	2026-09-15	Added I2C timing table